CMS Physics Analysis Summary

Contact: cms-pag-conveners-higgs@cern.ch 2014/07/05

Search for lepton flavor violating decays of the Higgs boson

The CMS Collaboration

Abstract

The first direct search for Lepton Flavor Violating Decays of the recently discovered Higgs boson using 19.7 fb^{-1} of $\sqrt{s} = 8$ TeV data taken in 2012 using the $H \to \mu\tau_e$ and $H \to \mu\tau_{had}$ channels is described , where τ_{had} and τ_e are taus reconstructed in the hadronic and electronic decay channels respectively. The sensitivity of the search is an order of magnitude better than the existing indirect limits. A slight excess of signal events with a significance of 2.5σ is observed. The local p-value of this excess at $M_H = 126$ GeV is 0.007. Interpreted as a limit this results in a constraint of $B(H \to \mu\tau) < 1.57\%$ at 95% confidence level. The best fit branching fraction is $B(H \to \mu\tau) = (0.89^{+0.40}_{-0.37})\%$. The limit is subsequently used to constrain the $Y_{\mu\tau}$ Yukawa coupling.

1 Theoretical Motivation

The discovery of the Higgs boson [1] has generated great interest in exploring its properties. In the standard model (SM) lepton flavor violating (LFV) decays are not allowed if the theory is to be renormalizable. If however, the requirement that it is renormalizable is relaxed, so that it is a theory valid only to a finite mass scale, then LFV couplings may be introduced. LFV decays can also occur naturally in models with more than one Higgs doublet without giving up on renormalizability [2]. They also arise in composite Higgs models [3, 4], models with flavor symmetries [5], Randall-Sundrum models [6] and many others.

The presence of LFV Higgs couplings would allow LFV effects in decays mediated by virtual Higgs. There are three possibilities $\mu \to e$, $\tau \to \mu$ and $\tau \to e$ transitions. The experimental constraints have been reviewed and translated into constraints on $B(H \to e\mu, \mu\tau, e\mu)$ in two recent papers [7, 8]. The $\mu \to e$ transition is strongly constrained by null searches for $\mu \to e\gamma$ [9], $B(H \to \mu e) < \mathcal{O}(10^{-8})$. The constraints on $\tau \to \mu$ and $\tau \to e$ are much less stringent. These come from searches for $\tau \to \mu\gamma$ and $\tau \to e\gamma$, muon and electron *g-2* measurements. Exclusion limits on their electron dipole moments also provide complementary constraints. These lead to the much less restrictive limits: $B(H \to \mu\tau) < \mathcal{O}(10\%)$, $B(H \to e\tau) < \mathcal{O}(10\%)$.

The observation of the Higgs boson offers the possibility of sensitive direct searches for the LFV Higgs decays $H \to \mu\tau$ and $H \to e\tau$. To date no dedicated searches have been performed. However in reference [8] the authors reinterpreted ATLAS $H \to \tau\tau$ search results in terms of LFV decays to set a 95% C.L of $B(H \to \mu\tau) < 13\%$, $B(H \to e\tau) < 13\%$. The possibility of LFV in Higgs decays up to about 10% of the rate motivates a dedicated search in CMS data.

2 Analysis Overview

A search for a LFV Higgs decay of a Higgs with $M_H = 126\,\text{GeV}$ [10] is performed in two channels $H \to \mu\tau_e$ and $H \to \mu\tau_{had}$, where τ_{had} and τ_e are taus reconstructed in the hadronic and electronic decay channels respectively. The 2012 dataset taken at $\sqrt{s} = 8$ TeV corresponding to an integrated luminosity of 19.7 fb^{-1} is used. The final state signature is very similar to the standard model $H \to \tau_\mu\tau_{had}$ and $H \to \tau_\mu\tau_e$ searches [11] but with some significant kinematic differences. The muon comes promptly from the LFV Higgs decay and tends to have a larger momentum than in the SM case, as shown in Figure 1. In addition, since the neutrinos in the decay arise from one tau which is heavily boosted, they tend to be collinear with the visible tau decay products.

The dominant background in the $H \to \mu\tau_e$ channel is $Z \to \tau_\mu\tau_e$. Other much smaller backgrounds come from jets faking leptons in W+jets, QCD multi-jets and $\text{t}\bar{\text{t}}$+jets events. In the $H \to \mu\tau_{had}$ channel the dominant background arises from jets faking taus in W+jets, QCD multi-jet and $\text{t}\bar{\text{t}}$+jets events. Less significant backgrounds also come from $Z \to \tau\tau$, $Z + jets$ and $\text{t}\bar{\text{t}}$+jets.

The tau leptons are reconstructed using the particle flow technique with the "hadrons plus strips" (HPS) tau-decay-mode finding algorithm [12]. Particle flow (PF) muons, electrons, jets and $E_{\text{T}}^{\text{miss}}$ are also employed. The two channels are divided into categories based on the number of jets in order to enhance the different Higgs production mechanisms. The dominant production mechanism is gluon-gluon fusion but there is also a significant contribution from vector boson fusion which is enhanced by requiring jets to be present in the event. The principle backgrounds are estimated using data. The presence or absence of signal is established using the asymptotic CL_s method. The analysis was performed "blinded" in the signal region. That is,

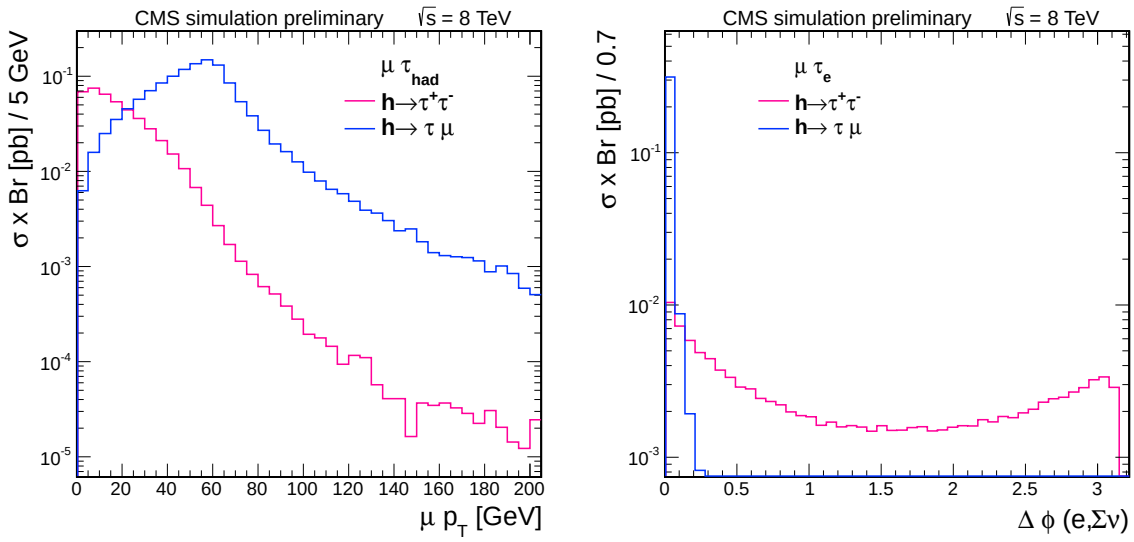

Figure 1: Comparison between simulated SM $H \to \tau\tau$ and LFV $H \to \mu\tau$ decays at the generator level. Left) p_T in the $\mu\tau_{had}$ channel. Right) Angle between the vector sum of the neutrinos in the event and the electron in the $\mu\tau_e$ channel.

the data in the signal region was not observed until the selection criteria had been fixed and the background estimate finalized.

3 Datasets

The LFV decay $H \to \mu\tau$ has been implemented in the recent version of PYTHIA [13]. The main Higgs production processes of gluon-gluon fusion and vector-boson fusion are taken into account and the full GEANT detector simulation is used. The dominant backgrounds, which arise from vector boson production, are simulated with either MADGRAPH [14] or PYTHIA.

The search is performed on the 2012, $\sqrt{s} = 8$ TeV dataset, comprising 19.7 fb^{-1} of data. The $H \to \mu\tau_{had}$ channel selection begins by requiring an isolated single muon trigger with a p_T^{μ} threshold of 24 GeV in the pseudo-rapidity range $|\eta| < 2.1$, while the $H \to \mu\tau_e$ channel requires a muon-electron trigger with p_T thresholds of 17 GeV ($|\eta| < 2.5$) and 8 GeV ($|\eta| < 2.4$), respectively. Loose electron and muon identification criteria and isolation are applied at the trigger level.

4 Event Reconstruction

A particle flow algorithm [15–17] combines the information from all CMS sub-detectors to identify and reconstruct the individual particles emerging from all vertices: charged hadrons, neutral hadrons, photons, muons, and electrons. These particles are then used to reconstruct the $E_{\mathrm{T}}^{\mathrm{miss}}$, jets, hadronic τ decays, and to quantify the isolation of leptons and photons.

The large number of proton interactions occurring per LHC bunch crossing (pileup), with an average of 9 in 2011 and 19 in 2012, makes the identification of the vertex corresponding to the hard-scattering process nontrivial. This affects most of the physics objects: jets, lepton isolation, etc. The tracking system is able to separate collision vertices as close as 0.5 mm along the beam direction [18]. For each vertex, the sum of the p_{T}^2 of all tracks associated with the vertex is

computed. The vertex for which this quantity is the largest is assumed to correspond to the hard-scattering process, and is referred to as the primary vertex in the event reconstruction.

Jets are reconstructed from all the PF particles using the anti-k_T jet algorithm [19] implemented in FASTJET [20], with a distance parameter of $R = 0.5$. The jet energy is corrected for the contribution of particles created in pileup interactions and in the underlying event. This contribution is calculated as the product of the jet area and an event-by-event transverse momentum density ρ, also obtained with FASTJET using all particles in the event. Charged hadrons, photons, electrons, and muons reconstructed by the PF algorithm have a calibrated momentum or energy scale. A residual calibration factor is applied to the jet energy to account for imperfections in the neutral-hadron calibration, the jet energy containment, and the estimation of the contributions from pileup and underlying-event particles. This factor, obtained from simulation, depends on the jet p_{T} and pseudo-rapidity η, and is on the order of 5% across the whole detector acceptance. Finally, a percent-level correction factor is applied to match the simulated jet energy response to the one observed. This correction factor and the jet energy scale uncertainty are extracted from a comparison between the observed data and simulation for γ+jets, Z+jets, and dijet events [21]. Particles from different pileup vertices can be clustered into a pileup jet, or significantly overlap a jet from the primary vertex below the p_{T} threshold applied in the analysis. Such jets are identified and removed using a boosted decision tree (BDT) [22] with the following input variables: momentum and spatial distribution of the jet particles, charged- and neutral-particle multiplicities, and compatibility of charged hadrons within the jet with the primary vertex. Jets originating from b-quark hadronization are identified using the combined secondary-vertex b-tagging algorithm [23].

Hadronically decaying τ leptons are reconstructed and identified using the HPS algorithm [12] which targets the main decay modes by selecting PF candidates with one charged hadron and up to two neutral pions, or with three charged hadrons. A photon from a neutral-pion decay can convert in the tracker material into an electron and a positron, which can then radiate bremsstrahlung photons. These particles give rise to several ECAL energy deposits at the same η value and separated in azimuthal angle, and are reconstructed as several photons by the PF algorithm. To increase the acceptance for such converted photons, the neutral pions are identified by clustering the reconstructed photons in narrow strips along the azimuthal (ϕ) direction.

5 Event Selection

The event selection consists of three steps. First a loose pre-selection defining the basic signature. The sample is then divided into categories. Finally requirements are placed on a set of kinematic variables designed to suppress the backgrounds.

The pre-selection for the $H \to \mu\tau_e$ sample requires an an isolated tight muon ($p_{\mathrm{T}} > 25$ GeV, $|\eta| < 2.1$) and an isolated tight electron ($p_{\mathrm{T}} > 10$ GeV, $|\eta| < 2.3$) of opposite charge lying within a region of the detector that allows good identification. The $H \to \mu\tau_{had}$ sample requires an isolated muon ($p_{\mathrm{T}} > 30$ GeV, $|\eta| < 2.1$) and a tightly isolated hadronic tau ($p_{\mathrm{T}} > 30$ GeV, $|\eta| < 2.3$) of opposite charge.

The events are then divided into categories within each sample according to the number of jets in the event. Jets are required to pass loose identification criteria, have $p_{\mathrm{T}} > 30$GeV and lie within the range $|\eta| < 4.7$. The zero jet category contains events primarily produced by gluon-gluon fusion. The one jet category contains events produced by gluon-gluon fusion but also events produced in association with a W or Z boson decaying hadronically. Events enhanced

in the vector boson fusion process are required to have two jets separated by a rapidity gap ($\Delta\eta > 3.5$) and to have an invariant jet-jet mass greater than 550 GeV. In the $H \to \mu\tau_e$ channel events in which at least one of the jets is identified as coming from a b-quark decay are vetoed.

Table 1: Selection criteria requirements for the kinematic variables after pre-selection.

Variable	$H \to \mu\tau_e$			$H \to \mu\tau_{had}$		
	0-jet	1-jet	2-jet	0-jet	1-jet	2-jet
$p_T^\mu >$ [GeV]	50	45	25	40	35	30
$p_T^e >$ [GeV]	10	10	10	-	-	-
$p_T^\tau >$ [GeV]	-	-	-	35	40	40
$\Delta\phi_{\vec{\mu}-\vec{\tau_{had}}} >$	-	-	-	2.7	-	-
$\Delta\phi_{\vec{e}-\vec{E}_T^{\text{miss}}} <$	0.5	0.5	0.3	-	-	-
$\Delta\phi_{\vec{e}-\vec{\mu}} >$	2.7	1.0	-	-	-	-
$M_T(e) <$ [GeV]	65	65	25	-	-	-
$M_T(\mu) >$ [GeV]	50	40	15	-	-	-
$M_T(\tau) <$ [GeV]	-	-	-	50	35	35

The signal variable is the collinear mass, $M_{collinear}$, which provides an estimator of the reconstructed Higgs mass using the observed decay products. This is constructed using the collinear approximation [24] which is based on the observation that since the mass of the Higgs is much greater than the mass of tau, the tau decay products are highly boosted in the direction of the original tau. Thus the neutrino momenta can be approximated to be in the same direction as the other visible decay products of the tau. Hence the component of the missing transverse energy in the direction of the visible tau decay products is used to estimate the transverse component of the neutrino momentum:

$$\vec{p}_T^{\nu} = \vec{E}_T^{\text{miss}} \cdot \hat{p}_T^{\tau_{vis}}$$

The fraction of the tau momentum carried by the visible tau decay products, $x_{\tau_{vis}}$, is given by

$$x_{\tau_{vis}} = \frac{|\vec{p}_T^{\tau_{vis}}|}{|\vec{p}_T^{\tau_{vis}}| + |\vec{p}_T^{\nu}|}$$

The tau four momentum is then $(x_{\tau_{vis}}|\vec{p}^{\tau_{vis}}|, x_{\tau_{vis}}\vec{p}^{\tau_{vis}})$ and since $M_H \gg m_\tau^2, m_l^2$

$$M_H = M_{collinear} = \frac{M_{vis}}{\sqrt{x_{\tau_{vis}}}}.$$

Figure 2 shows $M_{collinear}$ for the background regions compared to data for each of the categories in each channel after the pre-selection. The signal simulation for $B(H \to \mu\tau) = 100\%$ is shown. The principle backgrounds are estimated with data using techniques described in section 6. There is excellent agreement between data and the background estimation. The agreement is similarly good in all of the kinematic variables that are subsequently used to suppress backgrounds. The analysis was performed "blinded" in the region $100\,\text{GeV} < M_{collinear} < 150\,\text{GeV}$.

Next, a set of kinematic variables is defined and the criteria for selection is determined by optimizing for $S/\sqrt{S+B}$ where S and B are the expected signal and background event yields in the mass window $100\,\text{GeV} < M_{collinear} < 150\,\text{GeV}$. The signal strength is set according to the standard model Higgs production cross-section at $M_H = 126\text{GeV}$ with $B(H \to \mu\tau) = 10\%$. This value for the LFV Higgs branching fraction is chosen because it corresponds to the limit

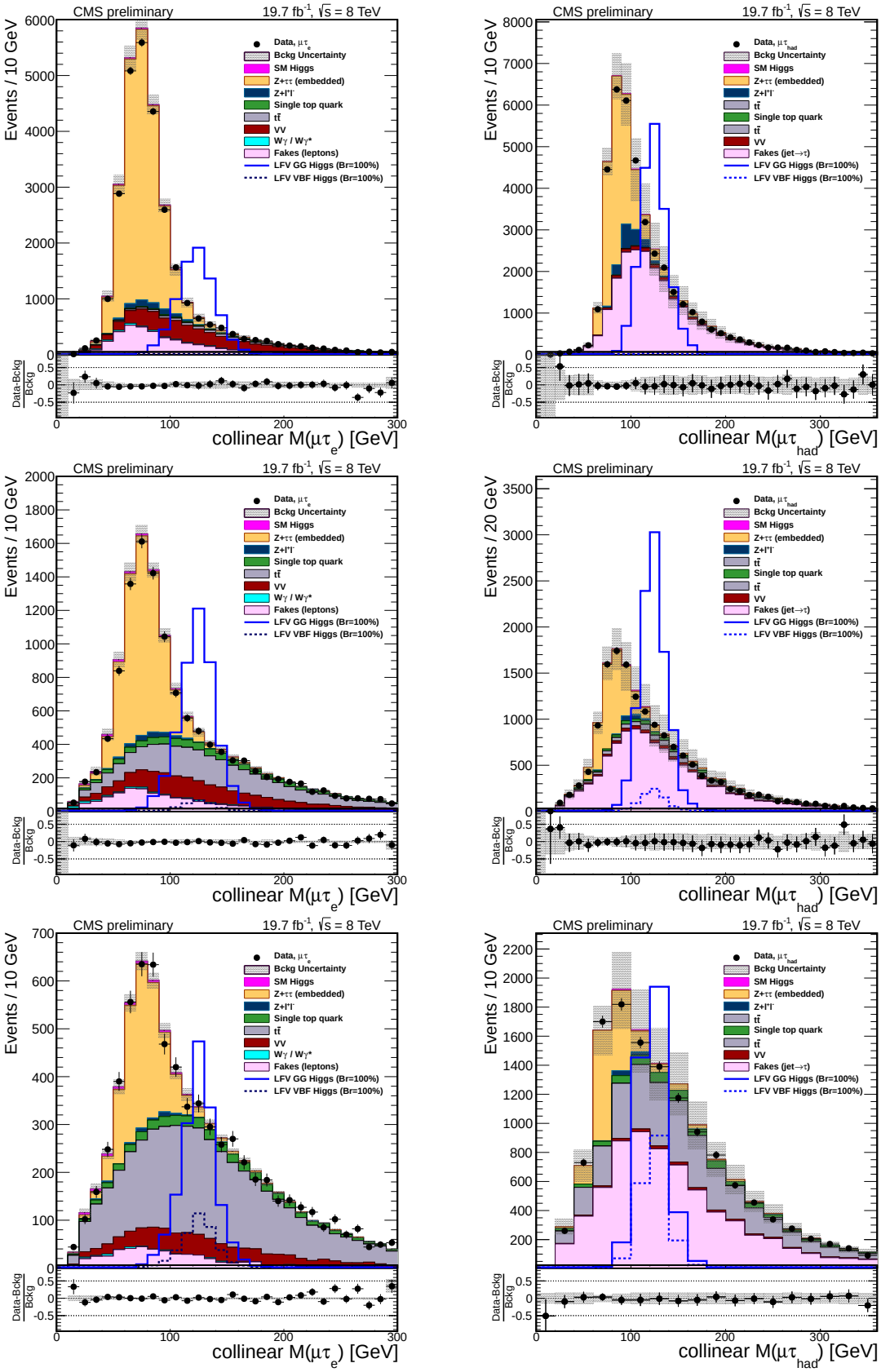

Figure 2: The collinear mass $M_{collinear}$ for signal ($B(H \to \mu\tau) = 100\%$ for clarity) and background after pre-selection requirements for the LFV $H \to \mu\tau$ decays for the different channels and categories compared to data. The shaded grey bands indicate the uncertainty. Top left: $H \to \mu\tau_e$ 0-jet , top right: $H \to \mu\tau_{had}$ 0-jet, middle left: $H \to \mu\tau_e$ 1-jet, middle right: $H \to \mu\tau_{had}$ 1-jet, bottom left: $H \to \mu\tau_e$ 2-jet, bottom right $H \to \mu\tau_{had}$ 2-jet.

from indirect measurements as described in reference [8]. The criteria for each category, and in each channel, are given in Table 1. The variables used are; the transverse momenta of the tau, muon and electron objects; azimuthal angles between the leptons, azimuthal angles between the leptons and the transverse missing energy vector, and finally the transverse mass. The transverse mass is constructed from the transverse missing energy vector $\vec{E}_{\mathrm{T}}^{\mathrm{miss}}$ and the lepton vector $\vec{l}$ as follows:

$$M_T(l) = \sqrt{2p_T(l)\vec{E}_{\mathrm{T}}^{\mathrm{miss}}(1 - \cos\Delta\phi_{\vec{l}-\vec{E}_{\mathrm{T}}^{\mathrm{miss}}})}$$

6 Backgrounds

The dominant backgrounds are estimated with data while the less significant backgrounds are estimated with simulations. The largest backgrounds come from $Z \to \tau\tau$ and from jets faking leptons in W+jets and QCD multi-jet events. The $Z \to \tau\tau$ background is estimated using an embedding technique in which simulated tau decays replace the muons in $Z \to \mu\mu$ data. The misidentified lepton rates are measured in an independent data set and then applied to control regions to estimate the background.

6.1 $Z \to \tau\tau$

The $Z \to \tau\tau$ background is estimated using a particle flow embedding technique. A sample of $Z \to \mu\mu$ events is taken from data using a loose selection. The muons are then replaced with simulated tau decays reconstructed with the particle flow algorithm. Thus the key features of the event topology such as the jets, missing energy and underlying event are taken directly from data with only the tau decays being simulated. The normalization of the sample is from the simulation expectation. The technique is validated by comparing identification efficiencies estimated with embedded decays to those from simulated $Z \to \tau\tau$ decays. It is found that the muon and hadronic tau efficiencies are well reproduced. There are small differences in the the electron identification efficiency. The differences are understood to be a consequence of the selective readout of the ECAL and corrections are applied for this.

6.2 Misidentified Leptons

Leptons can arise from misidentified jets from W+jets and QCD multi-jet events. This background is estimated using a data based method. It is employed slightly differently in the two channels. The technique is shown schematically in Table 2. The difference in the two channels is how the selection requirements are altered to define regions III and IV. In $H \to \mu\tau_e$ region I is the signal region in which an isolated muon and an isolated electron is required. Region III is a data sample in which all the analysis selection criteria are applied except that one of the leptons is required to be not-isolated, so that there are two components; isolated muon plus not-isolated electron events, and also isolated electron plus not-isolated muon events.

Table 2: Schematic to illustrate the application of the method used to estimate the misidentified lepton background. Samples are defined by the charge of the two leptons and by the selection requirements on each. The selection requirement and charge requirement is altered on the second lepton to define four regions.

	Opposite Sign leptons	Same Sign leptons
Selection(lepton1), Selection*(lepton2)	Region III	Region IV
Selection(lepton1), Selection(lepton2)	Region I	Region II

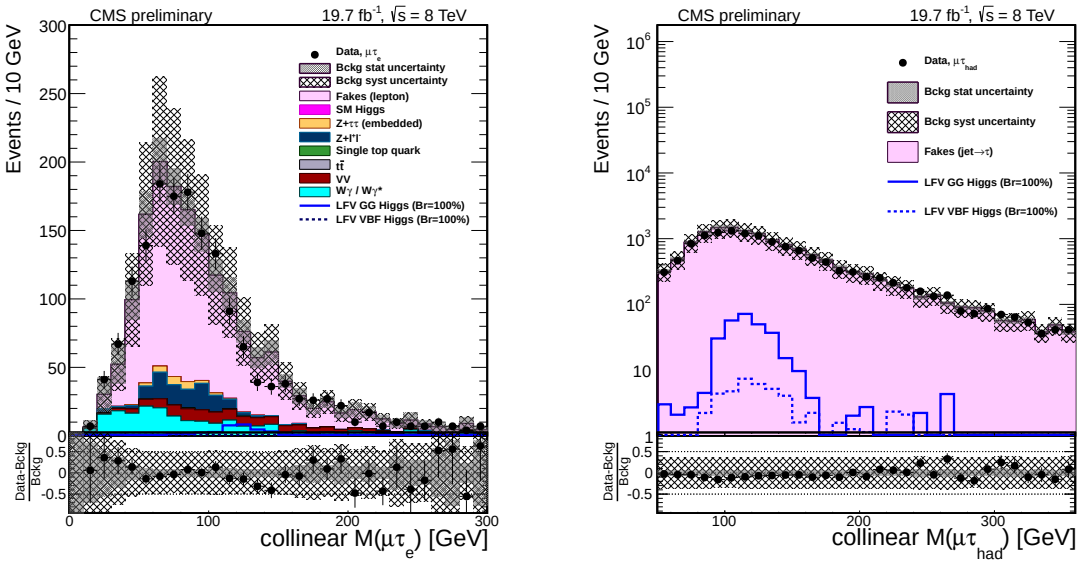

Figure 3: Left) $M_{collinear}$ for the $H \to \mu\tau_e$ same sign isolated e or μ plus isolated μ or e sample (region II). Right) $M_{collinear}$ for the $H \to \mu\tau_{had}$ same sign isolated muon plus tight-isolated tau (region II).

These samples are dominated by W+jets and QCD multi-jets but with small backgrounds from WW, ZZ that are subtracted using simulation expectations. The misidentified muon background in region I is then estimated by multiplying the event yield in region III by a factor $f_\mu \cdot \epsilon_{trigger}$, where f_μ is the ratio of not-isolated to isolated muons. It is computed in an independent data sample $\mathrm{Z}(\mu\mu) + (X = \mu)$

$$f_\mu = \frac{N_{events}(\mathrm{Z}(\mu\mu) + \mu_{isolated})}{N_{events}(\mathrm{Z}(\mu\mu) + \mu_{not-isolated})}.$$

in bins of muon p_T and η. A correction $\epsilon_{trigger}$ is made to account for the difference in trigger efficiency for selecting the isolated electron plus not-isolated muon versus selecting the isolated electron plus isolated muon. It is computed by selecting data in an independent single electron triggered data sample, applying the full event selection and then taking the ratio of the number of events that pass the electron-muon cross-trigger from the isolated versus not-isolated sample. The misidentified electron background is computed in exactly the same way. The technique is validated by using a like-sign rather than opposite-sign data sample in the same way as shown schematically in Table 2. Like sign leptons are chosen rather than opposite sign. Figure 3 shows the measured versus predicted by simulation $M_{collinear}$ distribution for the like sign control region. The agreement is very good.

In the $H \to \mu\tau_{had}$ channel tau leptons can be misidentified jets arising from a number of sources. Predominantly W+jets and QCD multi-jet events, but also $\mathrm{Z}(\mu\mu)$+jets , $\mathrm{t\bar{t}}$+jets. The misidentification rate f_τ is measured in $\mathrm{Z}(\mu\mu) + X = (\tau)$ events. It is defined as

$$f_\tau = \frac{N_{events}(\mathrm{Z}(\mu\mu) + X = (\tau, ID + tight - isolated))}{N_{events}(\mathrm{Z}(\mu\mu) + X = (\tau, ID + loose - isolated))}.$$

The denominator is the number of events with an object identified as a τ except that the requirement on isolation is loosened. The numerator is the number of events where the normal tau isolation (tight-isolated) requirement is made. The misidentification rate measured in $\mathrm{Z}(\mu\mu) + X = (\tau)$ data is checked by comparing to that measured in $\mathrm{Z}(\mu\mu) + X = (\tau)$ simulation and found to be in good agreement. The measured misidentification rate is then used

in the same way as previously described, and illustrated schematically in Figure 2. In this case region I, the signal region is the baseline event selection with both an isolated tau and an isolated muon. Region III comprises the baseline selection with an isolated muon and a *loose-isolated&not-tight-isolated* tau requirement. This region is dominated by W+jets and QCD multi-jet background and there good agreement between the data and the expected yields from simulation. The isolated muon plus *loose-isolated&not-tight-isolated* tau region is then used to estimate the misidentified lepton background using the following relation:

$$\frac{N_{events}(\tau_{tight-isolated})}{N_{events}(\tau_{loose-isolated.\&.not-tight-isolated})} = \frac{f_\tau N_{events}(\tau ID)}{(1-f_\tau)N_{events}(\tau ID)}.$$

Hence the background in the signal region (region I) is given by

$$N_{events}(\tau_{tight-isolated}) = \frac{f_\tau}{(1-f_\tau)} N_{events}(\tau_{loose-isolated.\&.not-tight-isolated}).$$

The procedure can be validated with like sign $\mu - \tau$ events. Figure 3 shows the excellent agreement between data and simulation for the like sign samples. As an additional test, the method is tested on opposite sign $-\tau$ events, with tight isolation applied to the tau, after the preselection and without the requirements on the kinematic variables. Again there is excellent agreement between data and simulation for this region.

The method assumes that the misidentification rate in $Z(\mu\mu) + X = (\tau)$ events is the same as for W+jets and QCD processes. To test this assumption the misidentification rates are measured in a QCD data control sample. They are found to be consistent, but slightly lower. A conservative 30% uncertainty is assigned. Finally as a cross-check the study has been performed also as a function of the number of jets in the event and similar agreement found.

6.3 Other Backgrounds

The leptonic decays of W's produced in top decays from $t\bar{t}$ pairs produces opposite sign dileptons and E_T^{miss}. This background is estimated using simulated $t\bar{t}$+jets events using the shape of the $M_{collinear}$ distribution from simulation and a data control region for normalization. The control region is the 2-jet selection but with the additional requirement that at least one of the jets is b-tagged in order to enhance the $t\bar{t}$ contribution.

Other smaller backgrounds come from WW, ZZ+jets,W+jets $+ \gamma$ and single top production. Each of these is estimated with simulation using both the shape and the normalization.

7 Systematic Uncertainties

The presence or absence of a signal is established using the asymptotic CL_s method [25]. Shape templates of $M_{collinear}$ distributions for the various background sources are used. The expected signal and background is then binned in $M_{collinear}$ and a likelihood technique is used to estimate the signal strength μ which in turn can be used to set confidence limits in the absence of a signal. The systematic uncertainties that are independent of $M_{collinear}$ are treated as "nuisance" parameters. There are also uncertainties that are dependent on $M_{collinear}$ which are estimated by changing the shape of the $M_{collinear}$ templates.

Table 3: Systematics. All systematics are treated as correlated between the categories, except where there are two numbers. In this case the number denoted with * is treated as uncorrelated between categories.

Systematic Uncertainty	$H \to \mu\tau_e$			$H \to \mu\tau_{had}$		
	0-jet	1-jet	2-jet	0-jet	1-jet	2-jet
electron trigger/ID/isolation	3%	3%	3%	-	-	-
muon trigger/ID/isolation	2%	2%	2%	2%	2%	2%
hadronic tau efficiency	-	-	-	9%	9%	9%
luminosity	2.6%	2.6%	2.6%	2.6%	2.6%	2.6%
$Z \to \tau\tau$ background	3+3*%	3+5*%	3+10*%	3+5*%	3+5*%	3+10*%
$Z \to \mu\mu, ee$ background	30%	30%	30%	30%	30%	30%
misidentified muon and electron background	40%	40%	40%	-	-	-
misidentified hadronic tau background	-	-	-	30+10*%	30%	30%
WW, ZZ+jets background	15%	15%	15%	15%	15%	65%
$t\bar{t}$+jets background	10 %	10 %	10+10*%	10 %	10 %	10+33*%
$W + \gamma$ background	100 %	100 %	100 %	-	-	-
B-tagging veto	3%	3%	3%	-	-	-
Single top production background	10 %	10 %	10 %	10 %	10 %	10%

7.1 $M_{collinear}$ independent systematics

The systematic errors are summarized in Tables 3 and 4. The uncertainty in the electron and muon selection (trigger, identification and isolation) are estimated using the tag and probe technique in $Z(\mu\mu, ee)$ data. The hadronic tau efficiency is estimated using tag and probe in $Z(\tau\tau)$ data. The uncertainty in the $Z(\tau\tau)$ background comes predominantly from the uncertainty in the tau efficiency. The uncertainties in the estimation of the misidentified lepton rate come from the difference in rates measured in different data samples (QCD multi-jet and W+jets). The uncertainty in the production cross-section of the backgrounds that have been estimated by simulation is also included.

There are several uncertainties that arise from the theoretical uncertainty in Higgs production cross-section which differ for each production mechanism contribution within each category. They are given in Table 4. Note that these enter both for the LFV Higgs and the SM Higgs background and are treated as 100% correlated.

Table 4: Theoretical Uncertainties in Higgs production.

Uncertainty	Gluon-Gluon Fusion			Vector Boson Fusion		
	0-jet	1-jet	2-jet	0-jet	1-jet	2-jet
parton density function	+9.7%	+9.7%	+9.7%	+ 3.6%	+3.6%	+3.6%
renormalization scale	+8 %	+10 %	-30%	+4 %	+1.5%	+2%
underlying event/parton shower	+4%	-5%	-10%	+10%	0%	-1%

7.2 $M_{collinear}$ dependent uncertainties

The shape uncertainties are summarized in Table 5.

For the embedded $Z(\tau\tau)$ $M_{collinear}$ distribution a 1% shift has been observed with respect to $Z(\tau\tau)$ simulations by comparing the means of both templates. This occurs only in the $H \to \mu\tau_e$ channel. The template has been corrected for this effect and a 100% uncertainty on this shift is used.

Table 5: Systematic uncertainties in the shape of the signal and background templates.

Systematic	$H \to \mu\tau_e$	$H \to \mu\tau_{had}$
Hadronic Tau energy scale	-	3%
Jet Energy scale	3-7%	3-7%
Unclustered energy scale	10%	10 %
Z($\tau\tau$) Bias	100%	-

The jet energy scale has been studied extensively by the CMS collaboration and a standard prescription is used in all analysis. The uncertainties are applied as a function of p_{T} and η including all correlations. The overall scale is set using $\gamma +$ jets events and the most significant uncertainty is in the photon energy scale. A number of other uncertainties such as jet fragmentation modeling, single pion response and uncertainties in the pileup corrections are also included. The jet energy scale uncertainties are propagated to the missing energy. There is also an additional uncertainty added to account for the unclustered energy scale uncertainty. The unclustered energy comes from jets below 10 GeV and particle flow candidates not within jets. It also is propagated to the missing energy.

The tau energy scale is estimated by comparing Z($\tau\tau$) events in data and simulation. Good agreement is found and a 3% uncertainty is assigned. Finally the shape templates have a bin by bin statistical uncertainty that is also included.

Potential uncertainties in the shape of the misidentified lepton backgrounds have also been considered. The procedure for estimating the misidentified lepton contributions is described in Section 6.2. In the $H \to \mu\tau_e$ channel the misidentified lepton rates are applied in bins of p_T and η. These rates are adjusted all up or down by 1 standard deviation and the different shapes are then used as nuisance parameters in the fit.

In the $H \to \mu\tau_{had}$ channel the tau lepton misidentification rate was found to be approximately flat in p_T and η. To estimate the systematic uncertainty the p_T distribution is fit with a linear function and the rate recomputed from the fitted slope and intercept. The modified shape template is then used to compute the systematic uncertainty.

8 Results

The post-fit $M_{collinear}$ distributions are shown in Figure 4 and the event yields in the mass range 100GeV $< M_{collinear} <$ 150GeV are shown in Table 6. The different channels and categories are used to set a 95% confidence level upper limit on the branching fraction of lepton flavor violating Higgs decay in $\mu\tau$ channel, $B(H \to \tau\mu)$, as described in the previous section.

The observed and the mean expected 95% CL upper limits for Higgs boson mass at 126 GeV are given for each category in Table 7. Combining all the channels an expected upper limit of $B(H \to \mu\tau) < (0.75 \pm 0.38)\%$ is obtained. The observed upper limit is $B(H \to \mu\tau) < 1.57\%$ which exceeds the expected limit due to a slight excess of observed number of events. The fit can then be used to estimate the branching fraction if this excess were to be interpreted as a signal.

The best fit values for the branching fractions are given in Table 7. The limits and best fit branching fractions are also summarized graphically in Figure 5. The combined categories give a best fit of $B(H \to \mu\tau) = (0.89^{+0.40}_{-0.37})\%$. The combined excess is 2.5 standard deviations σ which corresponds to a local p-value of 0.007 at $M_H = 126$ GeV.

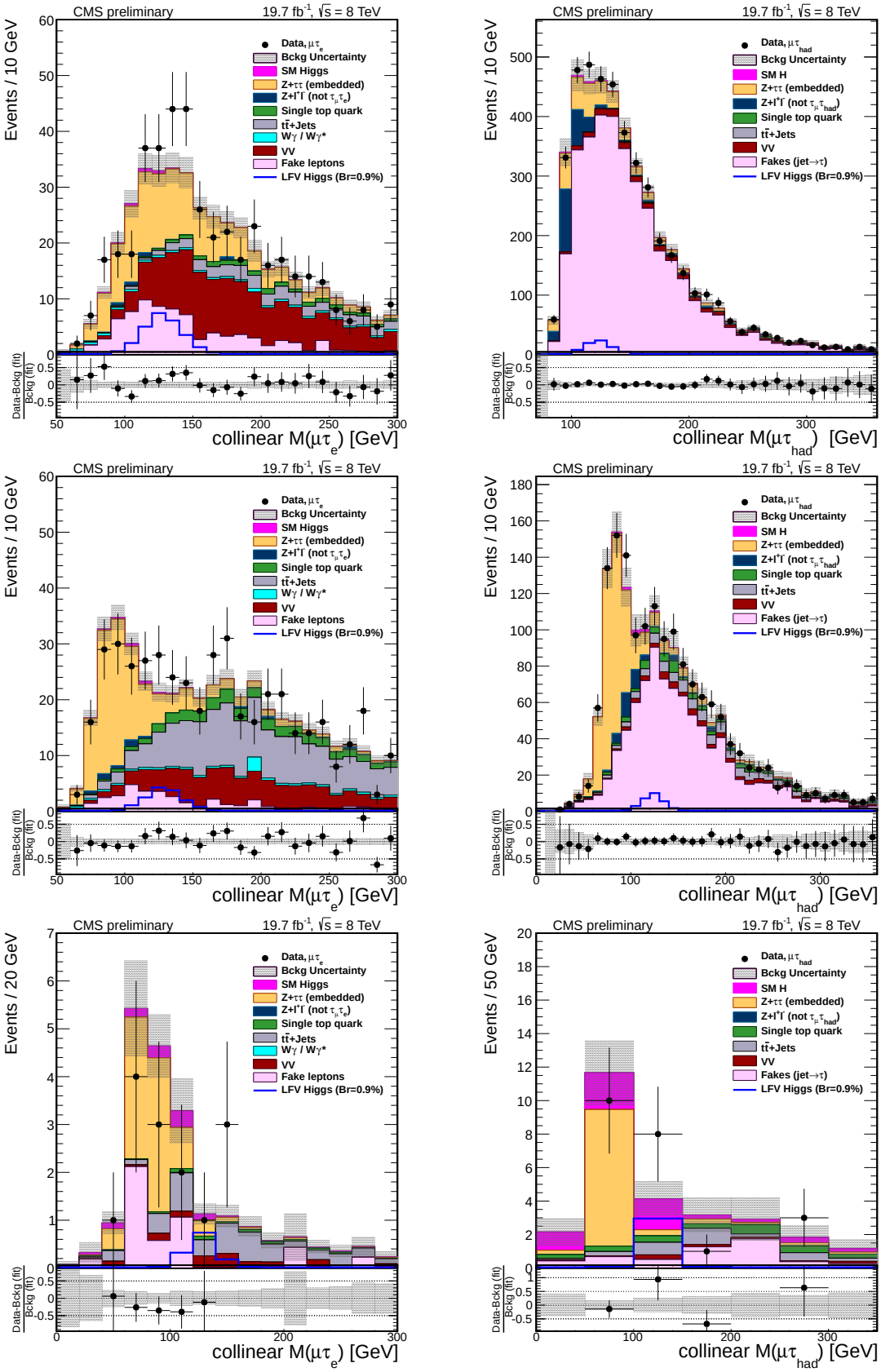

Figure 4: The collinear mass $M_{collinear}$ after fitting for signal and background for the LFV $H \to \mu\tau$ decays in the different channels and categories compared to data. top left: $H \to \mu\tau_e$ 0-Jet, top right: $H \to \mu\tau_{had}$ 0-Jet, middle left: $H \to \mu\tau_e$ 1-jet, middle right: $H \to \mu\tau_{had}$ 1-jet, bottom left: $H \to \mu\tau_e$ 2-jet, bottom right $H \to \mu\tau_{had}$ 2-jet.

Table 6: Event yields in the signal region, 100 GeV $< M_{collinear} <$ 150 GeV. The expected contributions are normalized to an integrated luminosity of 19.7fb^{-1}. The LFV Higgs signal is the simulation expection for $B(H \to \mu\tau) = 0.89\%$.

Sample	$H \to \mu\tau_{had}$			$H \to \mu\tau_e$		
	0-jet	1-jet	2-jet	0-jet	1-jet	2-jet
Fakes	1858.1 ± 558.8	362.9 ± 110.0	0.5 ± 0.5	41.5 ± 17.3	16.1 ± 6.8	1.1 ± 0.7
$Z \to \tau\tau$	198.8 ± 11.0	50.5 ± 3.5	0.4 ± 0.2	65.0 ± 3.0	38.6 ± 2.0	1.3 ± 0.2
ZZ, WW	47.0 ± 8.0	14.6 ± 2.6	0.3 ± 0.2	40.8 ± 6.6	21.2 ± 3.5	0.7 ± 0.2
$W\gamma$	–	–	–	2.0 ± 2.1	1.9 ± 1.9	–
$Z \to ee$ or $\mu\mu$	94.5 ± 25.2	17.6 ± 6.7	0.1 ± 0.1	1.6 ± 0.8	1.8 ± 0.8	–
$t\bar{t}$	2.5 ± 0.6	24.3 ± 3.2	0.7 ± 0.3	4.8 ± 0.7	30.0 ± 3.4	1.8 ± 0.3
$t, \bar{t}$	2.7 ± 1.2	19.9 ± 3.9	0.4 ± 0.5	1.9 ± 0.2	6.8 ± 0.8	0.2 ± 0.1
SM Higgs background	7.0 ± 1.3	4.9 ± 0.7	1.9 ± 0.7	1.9 ± 0.3	1.6 ± 0.2	0.6 ± 0.1
Sum of backgrounds	2210.4 ± 559.6	494.7 ± 110.4	4.3 ± 1.1	159.4 ± 18.9	118.1 ± 8.9	5.6 ± 0.9
LFV Higgs signal	69.7 ± 17.0	29.7 ± 6.7	3.0 ± 1.0	24.2 ± 5.7	13.6 ± 3.1	1.2 ± 0.4
data	2255.0 ± 47.5	506.0 ± 22.5	8.0 ± 2.8	180.0 ± 13.4	128.0 ± 11.3	6.0 ± 2.4

Table 7: The expected upper limits, observed limits and best fit values for the branching fractions for different jet categories for the $H \to \mu\tau$ process. The one standard-deviation probability intervals around the expected limit are shown in parentheses.

Expected limits			
	0 Jet (%)	1 Jet (%)	2 Jets (%)
$\mu\tau_e$	< 1.32 ($\pm$ 0.67)	< 1.66 ($\pm$ 0.85)	< 3.77 ($\pm$ 1.92)
$\mu\tau_{had}$	< 2.35 ($\pm$ 1.20)	< 2.10 ($\pm$ 1.07)	< 1.94 ($\pm$ 0.99)
$\mu\tau$	< 0.75 ($\pm$ 0.38)		
Observed limits			
$\mu\tau_e$	< 2.04	< 2.38	< 3.84
$\mu\tau_{had}$	< 2.94	< 2.11	< 3.29
$\mu\tau$	< 1.57		
Best fit branching fractions			
$\mu\tau_e$	$0.87^{+0.66}_{-0.62}$	$0.81^{+0.85}_{-0.78}$	$0.05^{+1.58}_{-0.97}$
$\mu\tau_{had}$	$0.72^{+1.18}_{-1.15}$	$0.03^{+1.07}_{-1.12}$	$1.24^{+1.09}_{-0.88}$
$\mu\tau$	$0.89^{+0.40}_{-0.37}$		

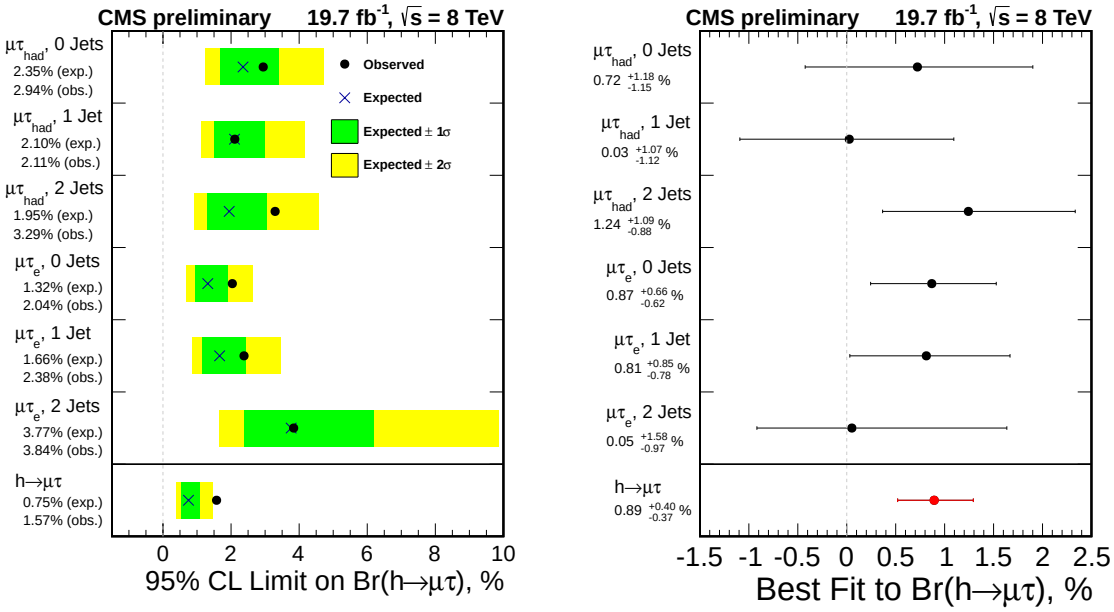

Figure 5: Left) Upper limits by category for the LFV $H \to \mu\tau$ decays. Right) Best fit branching fractions by category.

9 Extracting limits on lepton flavor violating couplings

The constraint on $B(H \to \mu\tau)$ can be interpreted in terms of LFV Higgs Yukawa couplings. The LFV decays $H \to e\mu, e\tau, \mu\tau$ arise at tree level from the assumed flavor violating Yukawa interactions where the relevant terms are explicitly

$$L_V \equiv -Y_{e\mu}\bar{e}_L\mu_R h - Y_{\mu e}\bar{\mu}_L e_R h - Y_{e\tau}\bar{e}_L\tau_R h - Y_{\tau e}\bar{\tau}_L e_R h - Y_{\mu\tau}\bar{\mu}_L\tau_R h - Y_{\tau\mu}\bar{\tau}_L\mu_R h$$

The branching fraction in terms of the Yukawa couplings are given by

$$B(H \to l^\alpha l^\beta) = \frac{\Gamma(H \to l^\alpha l^\beta)}{\Gamma(H \to l^\alpha l^\beta) + \Gamma_{SM}} \tag{1}$$

where $l^\alpha, l^\beta = e, \mu, \tau$ and $l^\alpha \neq l^\beta$. The decay width, in turn, is

$$\Gamma(H \to l^\alpha l^\beta) = \frac{m_h}{8\pi}(|Y_{l^\beta l^\alpha}|^2 + |Y_{l^\alpha l^\beta}|^2) \tag{2}$$

and SM Higgs width is $\Gamma_{SM} = 4.1$ MeV for a 125 GeV Higgs boson. It was assumed that at most one of non-standard decay mode of the Higgs is significant compared to the SM decay width.

The constraints on the Yukawa couplings derived from the limit $B(H \to \mu\tau) < 1.57\%$ are shown in Figure 6. This is compared to the constraints from previous indirect measurements. It can be seen that the direct search improves the constraint by roughly an order of magnitude.

10 Conclusions

The first direct search for lepton flavor violating decays of a Higgs boson to a muon-tau pair, based on the full 8 TeV dataset collected by CMS in 2012 is presented. The sensitivity of the

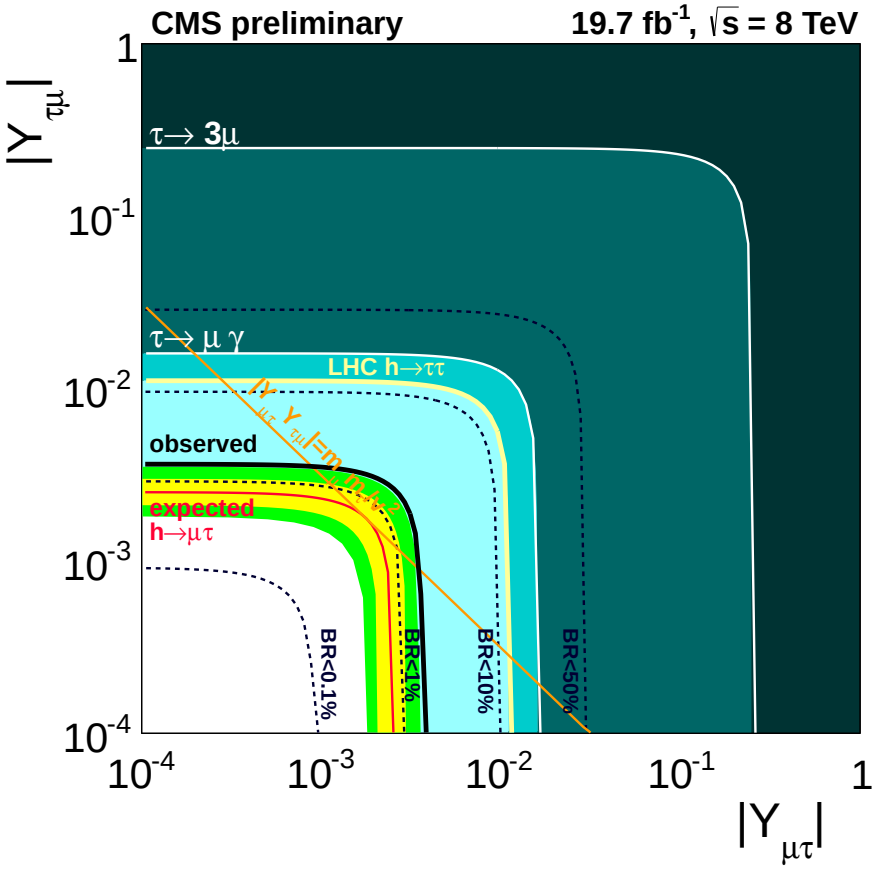

Figure 6: Constraints on the flavor violating Yukawa couplings, $|Y_{\mu\tau}|, |Y_{\tau\mu}|$. The expected (red solid line) and observed (black solid line) limits are derived from the limit on $B(H \to \mu\tau)$ from the present analysis. The diagonal Yukawa couplings are approximated by their SM values. The black dashed lines are contours of $B(H \to \mu\tau)$ for reference. The shaded regions are derived constraints from null searches for $\tau \to 3\mu$ (dark green) and $\tau \to \mu\gamma$ (lighter green). The orange diagonal line is the theoretical naturalness limit $Y_{ij}Y_{ji} \leq m_i m_j / v^2$. The yellow line is the limit from a reinterpretation, by a theoretical group [8], of an ATLAS $H \to \tau\tau$ search.

search is an order of magnitude better than the existing limits. A slight excess of signal events with a significance of 2.5σ is observed. Interpreted as a statistical fluctuation, this results in a constraint of $B(H \to \mu\tau) < 1.57\%$ at 95% confidence level. Interpreted as a signal, the best fit branching fraction is $B(H \to \mu\tau) = (0.89^{+0.40}_{-0.37})\%$. The limit is subsequently used to constrain the $Y_{\mu\tau}$ Yukawa coupling. It improves the current bound by an order of magnitude.

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